

Figure 1: The use of technology in education – various dimensions

in mind. By constantly improving its accessibility features, Sakai aims to comply with the Web Content Accessibility Guidelines (WCAG 2.0) Levels A and AA. Recent standards like WAI-ARIA are also taken into consideration.

- Sakai is extendable. Community-developed Sakai-specific innovations are termed as Contrib Tools. The integration of third party tools with Sakai should be done using LTI (Learning Tools Interoperability), which is a standard for secure integration.
- ePortfolio is another important feature of Sakai, which facilitates tools for exhibition, feedback and assessment. It should be used by students to showcase their accomplishments. The instructors should use ePortfolio for the formal evaluation of students' work.
- The Sakai project has support for various third party standards like xAPI (Experience API), Learning Information Services Specification (LIS), the Question and Test Interoperability Specification (QTI) and Sharable Content Object Reference Model (SCORM).

With this list of features, Sakai can make a great impact to the teaching-learning experience. The 'Teaching with Sakai Innovation Award' (TWSIA) has been established to recognise excellence in the teaching and learning process. For further information, check out <https://www.sakaiproject.org/teaching-sakai-innovation-award>.

BigBlueButton

BigBlueButton is an awesome tool for remote learning. It is open source with a very intuitive user interface. The development of BigBlueButton began in 2007 at Carleton University in Canada. The goal was to enable remotely located students to attend lecture sessions without having to travel. The most recent release of BigBlueButton is 0.9.0 (October 15, 2014).

The vision of BigBlueButton is to provide high quality online education to every student, with the only requirement being a Web browser on the student's computing device. The face-to-face communication between instructor and students is possible because of the Web conferencing facilities, as well as audio and video chat features. Document sharing is also enabled.

The desktop of the system can also be shared via BigBlueButton. Its features, listed in Figure 3, are as follows:

- It supports a variety of office document formats. PDF is



Figure 2: Sakai's features

considered to be the optimal format for BigBlueButton.

- Annotations can be added to the documents loaded in BigBlueButton. Apart from annotations, zoom in and out features are also supported.
- The sessions are recorded for playback later.
- Webcam-sharing is also supported by BigBlueButton, with no upper limit on the number of active Web cameras in use at the same time.

The students interact with their teachers on a real-time basis through BigBlueButton. The actions made at the teacher's system are immediately reflected in the students' systems linked to that session. The screen layout of the system can also be customised according to requirements, i.e., either the webcam or presentation space can be maximised.

Apart from the end-user features, BigBlueButton provides many developer-friendly features like APIs. The support for LTI (Learning Tool Integration) is another important feature.

Xerte

The most important element of education is the content. The multimedia properties of electronic content make it possible to create interesting learning sessions in an efficient manner. Xerte is an open source suite for creating interactive learning material. It is an excellent alternative to its commercial counterparts, which are much costlier. Being open source, Xerte is free and easier to use. It was developed by the University of Nottingham, UK.

Xerte is an awesome tool for creating learning objects that are a combination of good content, practice items and assessment items. Xerte packages are shown in Figure 4.

The Xerte online toolkit simply requires a Web browser

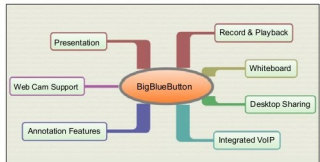


Figure 3: BigBlueButton's features